

DMAIC plus AI Master Cheat Sheet

The whole course on one sheet: the five DMAIC phases, what each is for, the key tools, and what AI does for you as a white belt in each phase.

Define: understand and scope the problem

Goal: agree on what problem you are solving, for whom, and what done looks like. Tools: project charter, problem statement, SIPOC, CTQs, Voice of the Customer.

AI for the white belt: translate jargon to plain English, decode a charter into a summary plus your role plus smart questions, organize a messy process brain-dump, and compile VOC by source (compile only, do not theme).

Measure: build a fact-based baseline

Goal: measure how the process performs today before changing anything. You cannot improve what you do not measure. Tools: process map, baseline data, histogram, run chart.

AI for the white belt: structure a process map (steps, owners, decision points, delays), bucket data into a histogram and describe its shape, and plot a run chart that flags a real signal versus normal noise.

Analyze: find the root cause

Goal: get past symptoms to the true cause. Tools: fishbone (Ishikawa) diagram with the six Ms, Five Whys, Pareto chart.

AI for the white belt: brainstorm a wide net of candidate causes across all six categories, run a Five Whys to drill toward the root, and categorize, count, and sort free-text notes into a Pareto that names the vital few.

Improve: design and pilot the fix

Goal: choose a solution that attacks the root cause, favors simplicity, and weighs side effects, then pilot before full rollout. Tools: divergent brainstorm, convergent stress-test, pilot test, SOP.

AI for the white belt: brainstorm solutions from simple to complex with effort estimates, play skeptic to expose how a fix could fail, and draft a clean SOP from a conversational description.

Control: lock in the gains

Goal: make the improvement stick and catch backsliding early. Tools: control chart (center line, UCL and LCL), standard work and SOP, response plan.

AI for the white belt: compute control-chart limits at three standard deviations and flag out-of-control points or runs, and draft a response plan covering who watches, what counts as a signal, and the first steps.

Your role: the white belt

- Local expert: you know how the work really happens.

- Data gatherer: you collect the raw, honest numbers.
- Documentation engine: you keep the charter, notes, and SOPs.
- Early-warning system: you watch the metrics and raise a flag when old habits creep back.

The LENS prompt recipe

Lead with the persona, Establish the context, Name the task, Set the format and constraints. A vague prompt gets a vague answer; LENS turns AI into a focused thinking partner.

The 95/5 rule: AI drafts, you verify

AI gets you about 95 percent of the way in under a minute: drafting, structuring, translating, brainstorming. You verify the 5 percent that matters: the facts, the numbers, the local truth, and the judgment calls. AI is great at drafting, mediocre at arithmetic and reading charts, and dangerously bad at strategic judgment. The machine brainstorms; the human decides.